

Password Strength Analyzer using Arduino

By Haya Fawaz Alswaied

Introduction:

This is a simple project that shows how to use the Arduino to check password strength, whether it is strong, medium, or weak, and then the LEDs will turn on to indicate the password strength: green, yellow, or red.

Required Components and Tools:

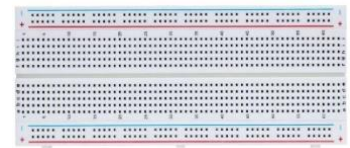
1-Arduino Uno



2- USB cable



3- Breadboard



4- Jumper wires

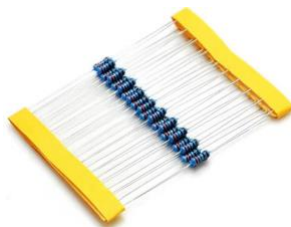
4 wires from each one



5- LEDs (green, yellow, red)



6- 3 Resistors (220Ω or 330Ω)



7- LCD Display (16x2 with I2C module)



8- Installed Arduino IDE

Arduino IDE download: arduino.cc/en/software/

This set includes all the required components and can be purchased from Amazon:
<https://amzn.eu/d/38rVzAU>

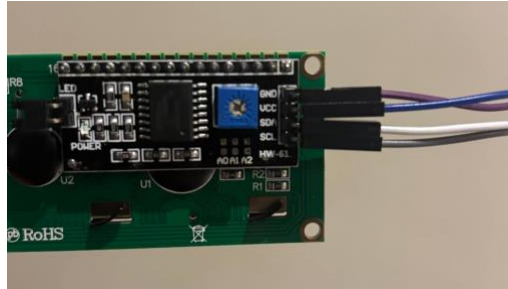
How to connect components with Arduino:

After purchasing all the components, we will learn, step by step, how to connect each component to the Arduino board.

1- Connect the LCD Display with the Arduino board

First, connect four wires to the LCD Display.

1. GND
2. VCC
3. SDA
4. SCL



then connect each wire from the LCD Display to the correct pin on the Arduino board.



LCD Display side	Arduino board side
GND	GND
VCC	5V
SDA	A4
SCL	A5

Connect the USB to your laptop.

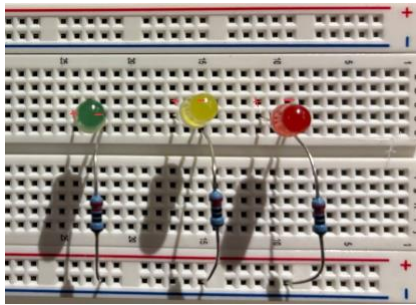


Finally, the LCD Display is connected to the Arduino board.

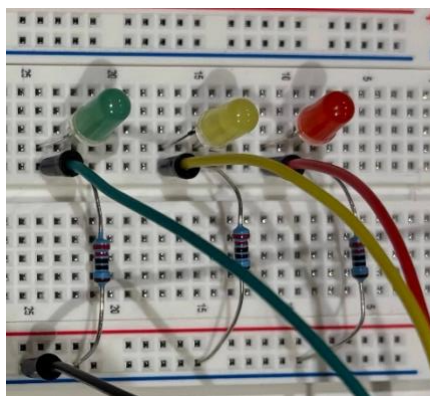


2- Connect the LEDs to the Arduino board

First, add LEDs and a resistors on the Breadboard as shown in the image.

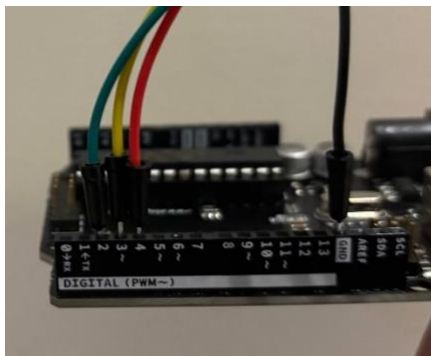


Then connect four wires on the Breadboard.



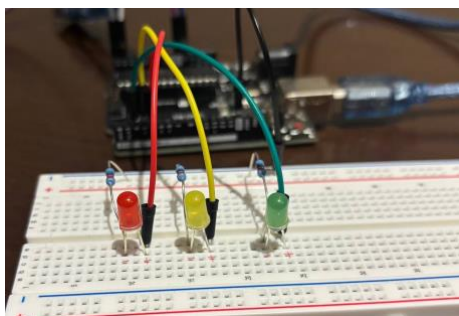
- 1- Black for GND
- 2- Green for green LED
- 3- Yellow for yellow LED
- 4- Red for red LED

Now connect each wire from the breadboard to the Arduino board.



Breadboard side	Arduino side
Black wire	GND
Green wire	2
Yellow wire	3
Red wire	4

Finally, everything is now connected to the Arduino board.



Arduino Code:

After connecting all components to the Arduino board, write the code below in the Arduino IDE.

Open the Serial Monitor from the Tools menu to enter a password. It will then show what is missing, such as length, uppercase and lowercase letters, digits, or symbols, so you know how to make the password stronger.

Before running the code, you need to install the LiquidCrystal_I2C library:

1. Click Sketch in the Arduino IDE.
2. Choose Include Library → Manage Libraries.
3. Search for LiquidCrystal_I2C.
4. Click Install.

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
String password = "";
```

```
int ledRed = 2;
```

```
int ledYellow = 3;
```

```
int ledGreen = 4;
```

```
void setup() {
```

```
  Serial.begin(9600);
```

```
  lcd.init();
```

```
  lcd.backlight();
```

```
  pinMode(ledRed, OUTPUT);
```

```
  pinMode(ledYellow, OUTPUT);
```

```
  pinMode(ledGreen, OUTPUT);
```

```
  lcd.setCursor(0,0);
```

```
  Serial.println("Enter your password:");
```

```
  lcd.setCursor(0,0);
```

```
  lcd.print("Enter Password: ");
```

```
}
```

```

void loop() {
    if (Serial.available() > 0) {
        password = Serial.readStringUntil('\n');
        bool check=checkPassword(password);
        if(!check) Serial.println("Enter another password:");
    }
}

```

```

bool checkPassword(String pwd) {
    bool hasUpper = false;
    bool hasLower = false;
    bool hasDigit = false;
    bool hasSymbol = false;
    bool passStrength =false;

    for (int i = 0; i < pwd.length(); i++) {
        char c = pwd.charAt(i);
        if (isUpperCase(c)) hasUpper = true;
        else if (isLowerCase(c)) hasLower = true;
        else if (isDigit(c)) hasDigit = true;
        else hasSymbol = true;
    }

    int score = 0;
    if (pwd.length() >= 8) score++;
    if (hasUpper) score++;
    if (hasLower) score++;
    if (hasDigit) score++;
    if (hasSymbol) score++;

    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print(pwd);
}

```

```

Serial.println("Password: "+ pwd);

lcd.setCursor(0,1);

if (score <= 2) {

    lcd.print("Weak Password!");

    Serial.println("Weak Password!");

} else if (score == 3 || score == 4) {

    lcd.print("Medium Password");

    Serial.println("Medium Password");

} else {

    lcd.print("Strong Password");

    Serial.println("Strong Password");

    passStrength=true;

}

showStrength(score);

if (pwd.length() < 8) Serial.println("Length less than 8");

if (!hasUpper) Serial.println("No uppercase letters");

if (!hasLower) Serial.println("No lowercase letters");

if (!hasDigit) Serial.println("No digits");

if (!hasSymbol) Serial.println("No special characters");

return passStrength;

}

void showStrength(int score) {

    digitalWrite(ledRed, LOW);

    digitalWrite(ledYellow, LOW);

    digitalWrite(ledGreen, LOW);

    if (score <= 2) {

        digitalWrite(ledRed, HIGH);

    } else if (score == 3 || score == 4) {

        digitalWrite(ledYellow, HIGH);

    } else {

        digitalWrite(ledGreen, HIGH);

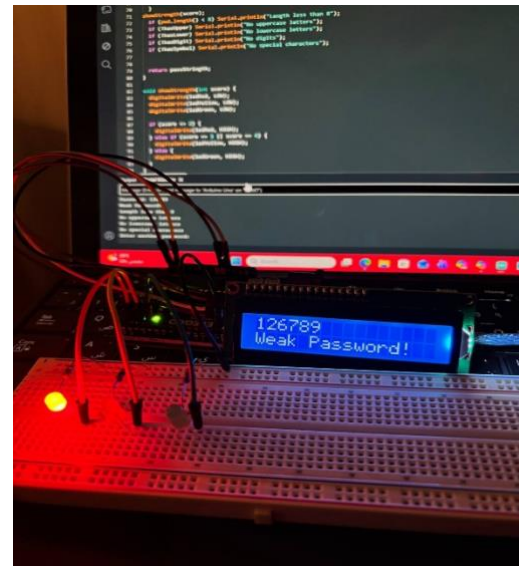
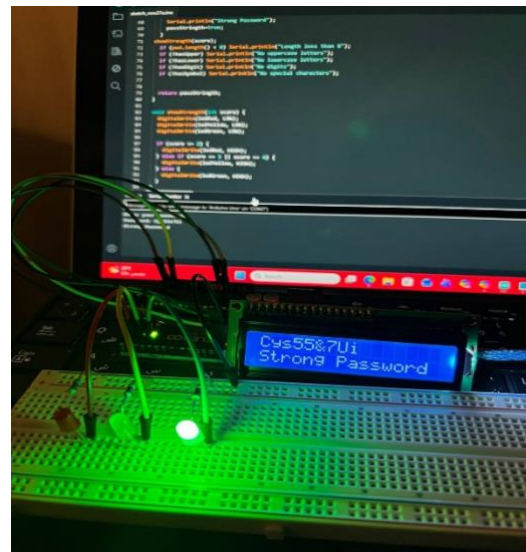
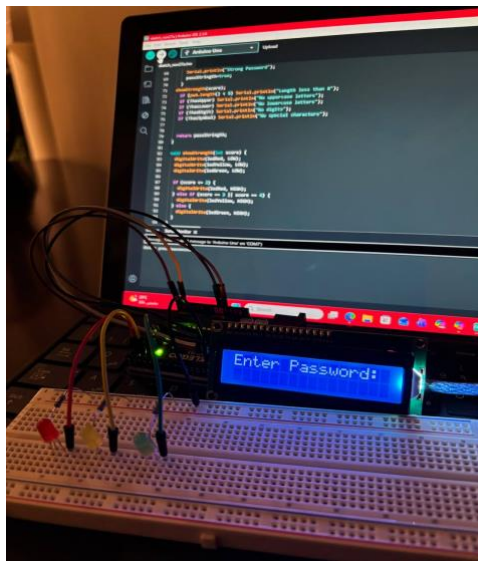
    }

}

}

```


Output:



```
Enter your password:  
Password: L1iiu670  
Medium Password  
No special characters  
Enter another password:
```

```
Password: 126789  
Weak Password!  
Length less than 8  
No uppercase letters  
No lowercase letters  
No special characters  
Enter another password:
```


Conclusion:

In conclusion, I listed all the components needed for this project and explained how to connect them to the Arduino board step by step. I hope the explanation was simple and clear, and that you enjoyed building and learning from this project. To improve the project, you can add a keypad to enter passwords directly without using the Serial Monitor. Another useful enhancement is to add a dictionary of common weak passwords and compare it with the user input, so the system can alert if the chosen password is too common or easy to guess.